

Hong Wenxuan

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EDUCATION

Bachelor of Engineering (Honours) in Information Engineering and Media Aug 2022 - Expected May 2026
Nanyang Technological University, Singapore

- Expected Second Upper Class Honours
- SG Digital Young Leader (IMDA)

PROFESSIONAL EXPERIENCE

Sectorial Risk Assessment Intern Jan 2025 – Aug 2025

Synapse, Sector Governance - Governance and Data Protection team

- Facilitated the rollout of an improved data security risk assessment methodology by engaging stakeholders, presenting updates, and addressing feedback to drive adoption across teams, while gaining hands-on experience in technical and process controls to ensure compliance with regulatory and organizational standards.
- Researched, drafted, and proposed requirements for a secure external file-sharing platform, evaluating solutions with data loss prevention, role-based access control, and other controls to mitigate risks.
- Assisted in planning and organizing internal cybersecurity forums, contributing to the enhancement of forum processes to improve efficiency.

COMPETITIONS AND PROJECTS

iNTUition v11.0 Hackathon Mar 2025

- Led the team to achieve 2nd runner-up in iNTUition v11.0 Hackathon (MSD track).
- Developed an AI assistant, Misty, in 24 hours, utilising OpenAI API and Retrieval Augmented Generation (RAG) to support change management professionals.
- Incorporated the ADKAR framework and analysed employee sentiment toward organizational change, enhancing communication and engagement through actional feedback.

DSTA BrainHack 2025 May 2025

- Competed in a 24-hour Capture the Flag (CTF) competition, solving 6 challenges and gaining hands-on experience with vulnerability identification and exploitation techniques.
- Utilized tools such as Wireshark for network analysis and FTK Imager for forensic investigation of leftover data artifacts on Kali Linux.
- Applied problem-solving strategies and time management in a team-based environment to tackle diverse cybersecurity challenges.

Final Year Project: Impact of Round Modifications on Cryptanalysis and Side-Channel Attacks in ASCON

Aug 2025 – Ongoing

- Investigating the impact of modifying round numbers and S-box structures on ASCON's resistance to linear cryptanalysis and vulnerability to side-channel attacks.
- Measuring outcomes via bias and key-recovery difficulty in linear cryptanalysis and Hamming weight and distance leakage models for side-channel vulnerabilities; analysing results to evaluate security implications.

Network Packet Sniffer using Python

July 2025 – Ongoing

- Built a raw socket-based packet sniffer on Windows to capture and decode live IPv4 network traffic including ICMP, TCP, UDP and DNS protocols.
- Extracted and displayed detailed packet header and payload information to support foundational network traffic analysis.

- Planning to integrate a user-friendly interface and AI-driven anomaly detection, similar to SIEM, to automatically identify outliers in network traffic.
- Reinforcement Learning: Evaluating Teacher’s Guidance on Agent Performance**

Oct 2024 – Dec 2024

 - Trained Reinforcement Learning Agents using Q-learning and SARSA, tuning hyperparameters and tracking performance over time in a simulated environment.
 - Implemented trained Q-learning and SARSA agents as teachers to guide new agent, evaluating impact of teacher availability and accuracy on the student agent’s average reward.
- Rescue Robot**

Aug 2024 – Dec 2024

 - Designed and built a remote-controlled rescue robot using Arduino Uno and MIT App Inventor, integrating an ESP32-CAM for live video streaming and ultrasonic sensors for distance detection.
 - Engineered a ball retrieval mechanism to simulate victim rescue, experimenting with various hardware configurations and optimising control algorithms for reliability and precision.

SKILLS

Programming Languages: Python, C, SQL, JavaScript, HTML
Technologies: Linux, Docker, Git, Google Cloud (Vertex AI, BigQuery ML)
Soft Skills: bilingual (Fluent in English and Chinese), teamwork, adaptability, leadership

LEADERSHIP

- Training and Development Sub-Committee Member**

June 2023 – June 2024

 - Facilitate school-wide workshops, including Figma and MATLAB workshops for over 200 participants, providing on-the-spot guidance, answering questions and assisting attendees throughout the sessions to ensure a smooth learning experience.
 - Collaborated with Google employees to organize and facilitate a week-long hackathon, assisting employees and participants throughout the event.

CERTIFICATIONS

Google Cybersecurity Professional Certificate	August 2025
Google IT Automation with Python	August 2025
Google Professional Machine Learning Engineer Certification	May 2025